off electrical devices based upon the nearby motion. You could also throw in some networking code, introduced in the last article, to create highly portable motion-based online IoT devices. Isn't this another strong reason to make the Super Mini a must-have part of your IoT toolkit?

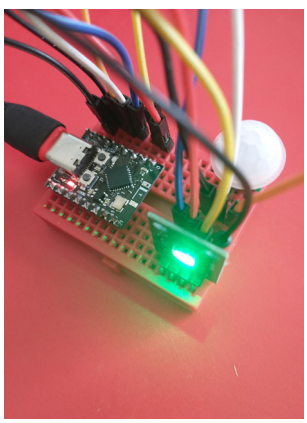


Figure 5: Mini PIR motion sensor with Super Mini

Our next sensor module to play with is an IR infrared obstacle avoidance sensor module. It has a pair of infrared transmitting and receiving sensors. When the transmitted light waves are reflected back, they will be received by the receiver. The onboard comparator circuitry does the processing and the green indicator LED comes to life. The simple code shown below is all that you require to start with the module. Please note the usage of Signal class to abstract the active low logic-based output from the sensor module.

```
#--- Start of obstklesnsr.
py ---
from machine import Pin,
soft_reset, Signal
from os import uname
from time import sleep

pfrm = uname()[-1].split()
[-1]
ahgh = True

eleds = (
          Signal(Pin(0, Pin.
OUT), invert=not(ahgh)),
          Signal(Pin(1, Pin.
OUT), invert=not(ahgh)),
          Signal(Pin(2, Pin.
OUT), invert=not(ahgh)),
        )

esnsr = Signal(Pin(5, Pin.
IN), invert=ahgh)

def aledoff():
  eleds[0].off()
  eleds[1].off()
  eleds[2].off()

def rledon():
  eleds[0].on()
  eleds[1].off()
  eleds[2].off()

def gledon():
  eleds[0].off()
  eleds[1].on()
  eleds[2].off()

def bledon():
  eleds[0].off()
  eleds[1].off()
  eleds[2].on()

aledoff()

try:
  while True:
    if esnsr.value():
      rledon()
      sleep(5)
    else:
      gledon()
      sleep(0.5)
except:
    aledoff()
    if -1 == pfrm.
find('ESP32C3'):
      soft_reset()

#--- End of obstklesnsr.
py ---
```

The breadboard assembly is shown in Figure 6. The module VCC, GND and OUT are connected to the Super Mini 3V3, GND and GPIO 5 pins respectively. After running the code, just create an obstacle using a finger in front of the sensor and you'll see the chip LED going Red for some time before turning back to Green. There is a preset on the sensor module which could be turned to set the range of obstacle detection.

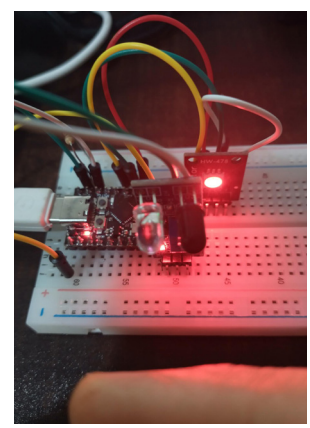


Figure 6: Obstacle avoidance sensor module with Super Mini

Last but not the least, there are a useful set of gas sensor modules like MQ-2 and MQ-135 to interface with the Super Mini. The MQ-2 gas sensor allows you to monitor LPG, smoke, and flammable gases. The MQ-135 is a popular metal oxide semiconductor gas sensor module. It detects and measures airborne contaminants like ammonia, nitrogen oxides (NOx), benzene, smoke, alcohol, and carbon dioxide. You could easily build a portable air quality device using these and the DHT temperature humidity sensor used in the last article. Just note, you have to connect MQ sensors VCC to the Super Mini 5V pin as those sensors use a small heater inside. You must also let the MQ sensors power up for a day to get them ready to provide correct output. You can add an inexpensive, easy-to-use 4-digit, 7-segment LED display module TM1637. Utilising the TM1637 driver IC, it requires only two data/clock pins to interface with the Super Mini, making it perfect for displaying sensors data.

The code given below is for a portable air quality monitoring device based upon DHT11, MQ-2/135 sensor modules, using the TM1637 library inline (link provided in the *Reference* section).

```
#--- Start of tempnumgss.
py ---
# include TM1637 library code
not shown here due to length

import dht
from machine import Pin, ADC,
Signal, reset
from os import uname
from time import sleep

pfrm = uname()[-1].split()
[-1].replace('-','')
if -1 != pfrm.find('ESP32C3'):
```